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# Atoms and isotopes

**Atomic structure**
Paper 1 · Combined and Separate
15 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 4 starred (★) questions first.

| | QUESTION | ANSWER | ✓ |
|---|---|---|---|
| 1 | **State the approximate radius of an atom.** | About 0.1 nm ($1 \times 10^{-10}$ m) | ☐ |
| 2 ★ | **Describe the structure of an atom.** | A positive nucleus of protons and neutrons, surrounded by negative electrons. | ☐ |
| 3 ★ | **State how the radius of a nucleus compares with the radius of an atom.** | Less than 1/10 000 of the radius of the atom. | ☐ |
| 4 | **State where most of the mass of an atom is concentrated.** | In the nucleus. | ☐ |
| 5 | **State the sign of the overall charge of the nucleus.** | Positive | ☐ |
| 6 | **State the relative charge of a proton.** | +1 (positive) | ☐ |
| 7 | **State the relative charge of a neutron.** | 0 | ☐ |
| 8 | **State the relative charge of an electron.** | −1 (negative) | ☐ |
| 9 | **State the relative mass of a proton.** | 1 | ☐ |
| 10 | **State the relative mass of a neutron.** | 1 | ☐ |
| 11 | **State the relative mass of an electron.** | Very small (about 1/2000) | ☐ |
| 12 | **Describe the arrangement of electrons in an atom.** | In energy levels at different distances from the nucleus. | ☐ |
| 13 ★ | **State how an electron can move to a higher energy level.** | By absorbing electromagnetic radiation. | ☐ |
| 14 ★ | **State how an electron can move to a lower energy level.** | By emitting electromagnetic radiation. | ☐ |
| 15 | **Name the two types of particle found in the nucleus of an atom.** | Protons and neutrons | ☐ |

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